Course Information

Course Number and Name

ECE 36200 Microprocessor Systems and Interfacing

Course Reference Number (CRN)

- Lecture: 17750
- Labs: 17752–17759 (Sec 003–010) and 59581 (Sec 011)

Instructional Modality: Face-to-Face

Course Credit Hours: 4

Prerequisites: ECE 26400 and ECE 27000 and (ENGR 11700 or CS 15900)

Course Description

An introduction to basic computer organization, microprocessor instruction sets, assembly language programming, and microcontroller peripherals.

Learning Resources, Technology & Texts

Textbooks

- (Highly recommended) Computer Organization and Design RISC-V Edition: The Hardware Software Interface, David Patterson, and John Hennessey (2nd Edition), ISBN-13: 9780128245583
- (Optional) Embedded Systems with ARM Cortex-M Microcontrollers in Assembly Language and C (3rd Ed.), Yifeng Zhu, ISBN-13: 978-0982692660

Lab Kit

- ECE 362 Lab Kit (purchase link)
- Breadboard and wire kit (can reuse ones from a previous course from a previous course, or you may purchase these ones from Amazon: breadboard and wire kit)

iClicker

- iClicker Student App (download link)

Webpages

- Brightspace: Lecture content delivery (link)
- Piazza: Discussions and Q&A (link)
- Gradescope: Homework and lab submission (link)

Learning Outcomes

A student who successfully fulfills the course requirements will have demonstrated:

- An ability to program a computer using assembly language.
- An ability to use a computer to interface with other devices.
- An ability to understand and simulate the functionality of a single-cycle processor.
- An understanding of basic computer arithmetic algorithms.

Grading Policy

Grade Breakdown

Component	Weight	Note
Labs	31%	Lab 0: 1%; Labs 1–10: 3% each
Project	16%	
Midterm	15%	
Final	20%	
Homework	10%	Roughly 1 per week. Lowest 1 dropped.
Embedded Practical	8%	
Total	100%	
Bonus: Participation	2%	See “Attendance Policy”
Bonus: PCB design	4%	Details announced with project.
Bonus: Spark Challenge	1%	Details announced with project.

Breakdown subject to change. Any changes will be posted in Brightspace and Piazza.

Grade Scale

Grade	Maximum Cutoff (*)
A+	97
A	93
A-	90
B+	87
B	83
B-	80
C+	77
C	73
C-	70
D+	67
D	63
D-	60
F (**)	Below 60

(*) The listed cutoffs represent the maximum thresholds. Final cutoffs may be the same or lower at the Instructor’s discretion.

(**) Regardless of your score, an "F" grade will be assigned if you:

- fail to take the midterm, final, and/or embedded practical without prior arrangement with the Instructor,
- fail to submit at least 50% of homework assignments, or
- fail to complete at least 50% of labs.

Requests for individual grade adjustments will not be considered.

Attendance Policy

In-person attendance is essential for your best learning experience. Attendance will be recorded through participation in daily pop quizzes using iClicker. Pop quizzes may be given at any point during the lecture and may not be given in some lectures.

Your participation will be recorded, and your participation bonus will be calculated:

$$\text{Bonus} = \min(2, \text{your_participiated_quiz} / (\text{total_quiz} - 6) \times 2)$$

You may miss up to six (6) lectures without losing any participation bonus. If you miss a lecture for any reason (illness, conference, etc.), please do not request a make-up. For extended absences, you must provide documentation to the Instructor.

Course Schedule

Week	Note	Lecture	Lab
Week 1		Introduction, GPIO	Lab 0: Introduction
Week 2	Labor Day	Interrupts, Timers	Lab 1: GPIO
Week 3		DMA, ADC/DAC	Lab 2: Interrupts
Week 4		ADC/DAC, PWM	Lab 3: Timers
Week 5		Serial communication	Lab 4: ADC, DMA
Week 6		Serial communication	Lab 5: PWM
Week 7			Lab 6: SPI, Project Proposal Deadline
Week 8	Fall Break, Midterm Review	Computer abstraction	No lab
Week 9	Midterm	Basic ISA	Lab 7: UART
Week 10		Control flow	Embedded Practical
Week 11		Procedures	Lab 8: Intro to RISC-V Assembly
Week 12		Single-cycle processor	Lab 9: Control Flow
Week 13		Single-cycle processor	Lab 10: Functions
Week 14	Thanksgiving	Arithmetic units	No Lab
Week 15	Final Review	Arithmetic units, Floating points	Project Deadline
Week 16	Quiet Week		
Week 17	Final (TBD)		

Schedule subject to change. Any changes will be posted in the learning management system. Refer to Academic Calendar for key University dates for the semester.

Examinations Policy

There will be one midterm exam and one final exam.

Exam Format

- Both exams will be pen-and-paper exams.
- Exams are expected to include a combination of multiple choice, multiple select, short answer, and/or free-response (workout) problems.
- Most questions will be automatically graded by Gradescope. No individual partial credit requests will be considered for the automatically graded questions.

Exam Schedule

- The midterm will be administered in Week 9, and the final will be administered in Week 17 (the finals week), though this schedule is subject to change. A 2-hour evening exam will be scheduled for each exam. Exact date and location will be posted at least one week before each exam.
- Students must ensure they are available during these scheduled times. No exceptions or rescheduling will be permitted under any circumstances, with the following exceptions:
    - Medical illness or emergency. Requires official documentation from a licensed US healthcare provider.
    - Death of an immediate family member or close relation. Requires coordination through the Office of the Dean of Students.
    - Participation in official Purdue-sponsored activities (e.g., ROTC, intercollegiate athletics, academic conferences, or study abroad programs).

  Students must present documentation confirming that the excuse applies specifically to the exam period. If approved, a make-up examination or alternative assessment arrangement may be offered at the Instructor's discretion. Excuses that will not be considered include but are not limited to: personal or leisure travel, internship obligations, employment conflicts, or other discretionary commitments. Makeup requests will not be granted for these reasons, so avoid scheduling potentially conflicting commitments until the exam dates are formally announced.
- Accommodated testing will be scheduled and provided through Purdue Testing Services (PTS) for DRC-eligible students.

Homework Policy

- Homework assignments will be administered throughout the semester, with one assignment due weekly.
- The highest N-1 homework scores will be used in calculating the homework portion of the final course grade, where N is the number of total homework assignments.
- Homework assignments will be posted in Gradescope.
- Homework submitted within one week after the deadline will receive a 20% penalty. Homework submitted more than one week late will receive a grade of zero. Deadline extension will not be considered under any circumstances.

Lab Policy

Refer to the lab policy document available on Brightspace.

Project

The course project is designed to provide hands-on experience in embedded system design. Students will work in teams to propose, design, implement, and test a microcontroller-based embedded system using the Purdue Proton board and other approved hardware/software resources.

Specific project documents will be made available on Brightspace.

Regrade Requests Policy

Homework and exam grades will be posted on Gradescope. Gradescope automatically enables regrade requests by default. Students may submit regrade requests within seven (7) calendar days of the grade being posted. After this window, regrade requests will not be accepted. It is the student's responsibility to review their graded work promptly and submit any regrade requests within the allowed period. All regrade requests must be specific, respectful, and clearly explain the grading concern. Requests submitted without a valid explanation or made without reasonable expectation of merit will not be considered.

Academic Integrity Policy

Academic integrity is a core value of Purdue University and is fundamental to the mission of learning, discovery, and engagement. All members of the Purdue community are expected to uphold the highest standards of academic conduct. Violations of this standard will not be tolerated and will be subject to disciplinary action.

Examples of academic dishonesty include but are not limited to:

- submitting homework that is not your own work, including solutions generated by AI tools such as ChatGPT or obtained from other students
- sharing exam content or answers with others during or after an examination
- accessing unauthorized websites or resources (e.g., Chegg or similar platforms) during an exam, regardless of whether assistance was obtained
- continuing to write or work on an exam after the allowed time has ended
- attempting to tamper with, hack, or manipulate class-related electronic systems or platforms
- submitting falsified or altered documentation as justification for an excuse
- requesting a regrade on an exam or assignment that has been altered after grading

Consequences for academic dishonesty may include:

- receiving a grade of zero (0) on the assignment, lab, or exam in question
- earning a failing grade for the course

The Instructor will follow the University's standard procedures for ensuring academic integrity, including reporting all incidents to the Office of Student Rights and Responsibilities (OSRR). The determination of penalties will be at the discretion of the Instructor, in accordance with University guidelines. Refer to the Academic Integrity document at https://www.purdue.edu/odos/osrr/academic-integrity/index.php.

Individuals are encouraged to alert university officials to potential breaches of this value by either emailing integrity@purdue.edu or by calling 765-494-8778. While information may be submitted anonymously, the more information is submitted the greater the opportunity for the university to investigate the concern.

AI Policy

Students are explicitly prohibited from presenting the output of such tools (e.g., ChatGPT's response to any prompt) as their own work in any type of course assignment. Such an action will be treated as an academic integrity violation, and will be handled the same way as any other academic integrity violation. When in doubt, always ask a member of course staff. For the purposes of assigned work that is intended to be created by you (i.e., labs, homeworks, etc.) we are treating LLMs like a person. If it would be an academic violation to ask your friend and take their response, then it's a violation to take an LLM's response.

Nondiscrimination Statement

Purdue University is committed to maintaining a community that recognizes and values the inherent worth and dignity of every person; fosters tolerance, sensitivity, understanding, and mutual respect among its members; and encourages each individual to strive to reach his or her potential. Please review the Purdue's Nondiscrimination Policy Statement at https://www.purdue.edu/home/ea_eou_statement/.

Accessibility

Purdue University strives to make learning experiences accessible to all participants. If you anticipate or experience physical or academic barriers based on disability, you are encouraged to contact the Disability Resource Center at: drc@purdue.edu or by phone: 765-494-1247, as soon as possible.

If the Disability Resource Center (DRC) has determined reasonable accommodations that you would like to utilize in my class, you must release your Course Accommodation Letter to me. Instructions on sharing your Course Accommodation

Letter can be found by visiting: https://www.purdue.edu/drc/students/course-accommodation-letter.php Additionally, you are strongly encouraged to contact me as soon as possible to discuss implementation of your accommodations.

Accommodated Testing

Students with disabilities whose DRC Course Accommodation Letter (CAL) includes test accommodations must first release their CAL to me and then schedule to take their exams with TPS at https://www.purdue.edu/studentsuccess/testing-services/accommodated-testing/student.php. You must schedule at least four days (96 hours) before the exam date listed on the syllabus.

In the case of finals week, you must schedule by the Friday before quiet period. The Instructor will provide TPS with your exam, and they will proctor it and provide the result to me for grading. Students must inform the Instructor immediately of cases when PTS is at capacity or otherwise unable to proctor the exam so that the Instructor can make other arrangements. Students who fail to follow this process and meet stated deadlines risk not being able to have their accommodations for that exam.

Mental Health/Wellness Statement

If you find yourself beginning to feel some stress, anxiety and/or feeling slightly overwhelmed, try Therapy Assistance Online (TAO), a web and app-based mental health resource available courtesy of Purdue Counseling and Psychological Services (CAPS). TAO is available to all students at any time by creating an account on the TAO Connect website, or downloading the app from the App Store or Google Play. It offers free, confidential well-being resources through a self-guided program informed by psychotherapy research and strategies that may aid in overcoming anxiety, depression and other concerns. It provides accessible and effective resources including short videos, brief exercises, and self-reflection tools.

If you need support and information about options and resources, please contact or see the Office of the Dean of Students. Call 765-494-1747. Hours of operation are M–F, 8am–5pm.

If you find yourself struggling to find a healthy balance between academics, social life, stress, etc., sign up for free one-on-one virtual or in-person sessions in West Lafayette with a Purdue Wellness Coach at RecWell. Student coaches can help you navigate through barriers and challenges toward your goals throughout the semester. Sign up is free and can be done on BoilerConnect. Students in Indianapolis will find support services curated on the Vice Provost for Student Life website.

If you are struggling and need mental health services: Purdue University is committed to advancing the mental health and well-being of its students. If you or someone you know is feeling overwhelmed, depressed, and/or in need of mental health support, services are available. For help, such individuals should contact Counseling and Psychological Services (CAPS) at 765-494-6995 during and after hours, on weekends and holidays, or by going to the CAPS offices in West Lafayette or Indianapolis.

Emergency Preparedness

In the event of a major campus emergency, course requirements, deadlines and grading percentages are subject to changes that may be necessitated by a revised semester calendar or other circumstances beyond the Instructor's control. Relevant changes to this course will be posted onto the course website or can be obtained by contacting the Instructors or TAs via email or phone. You are expected to read your @purdue.edu email on a frequent basis.

See Purdue's Information on Emergency Preparation and Planning. This website covers topics such as Severe Weather Guidance, Emergency Plans, and a place to sign up for the Emergency Warning Notification System. Students are encouraged you to download and review the Emergency Preparedness for Classrooms document.